MILLIY SERTIFIKAT MOCK TEST SAVOLLARI

129-MILLIY SERTIFIKAT TESTI

1. Agar $a = 4 \cdot 3^{m+2}$ va $b = 12^m$ sonlarining $EKUB(a; b) = 108$ bo'lsa, a ni toping.

A) 972 B) 729 C) 243 D) 324

2. 17^{36} sonini 5 ga bo'lgandagi qoldiq bilan 3333^{5555} sonining oxirgi raqami yig'indisini toping.

A) 5 B) 6 C) 7 D) 8

3. Hovuzdagi suv chiqarila boshlanganidan 2 soat o'tgach, unda $460\ m^3$ suv qoldi. Yana 3 soat o'tgach esa hovuzda $280\ m^3$ suv qoldi. Suv bir xil tezlikda chiqarilgan bo'lsa, dastlab hovuzda qancha suv bo'lgan?

A) $520\ m^3$ B) $560\ m^3$ C) $580\ m^3$ D) $640\ m^3$

4. Sotuvchi 1 kg bodomni 40 000 so'mdan sotib oldi. Bodom tozalanganda uning og'irligining 20% qismi chiqindi bo'lib ketdi. Agar sotuvchi 25% foyda ko'rmoqchi bo'lsa, tozalangan bodomning 1 kilogrammini necha so'mdan sotishi kerak?

A) 50 000 so'm B) 60 000 so'm C) 62 500 so'm D) 64 000 so'm

5. Ifodani soddalashtiring.

$$\frac{50^2 \cdot 250^{12} \cdot 625^{-4} \cdot 81^6}{4^3 \cdot 80^{14}}$$

A) 5^5 B) 5^{10} C) 10^{10} D) 10^{15}

17. Tenglamaning butun yechimlari yig'indisini toping. (Agar u bitta bo'lsa shu ildizini toping).

$$1 + \frac{6}{x-1} = \frac{5-2x}{x-7} + \frac{6(2x-5)}{x^2-8x+7} + \frac{1}{x-4}$$

A) 7 B) 8 C) 0 D) 6

18. Tengsizlikni yeching.

$$\frac{1}{(x-4)^2} + \frac{1}{x-4} - 20 \geq 0$$

A) $[3, 8; 5) \cup (5; 6, 25]$ B) $[9; 12) \cup (12; 16]$
C) $[2, 8; 5) \cup (6; 8]$ D) $[3, 8; 4) \cup (4; 4, 25]$

19. Tengsizlikning natural yechimlari yig'indisini toping.

$$(x^2 - 18x + 77)\sqrt{10 - x} \geq 0$$

A) 24 B) 38 C) 45 D) 50

20. Agar $(x^2 - 2x + m)^2 - 20$ ifodaning eng kichik qiymati -11 ga teng bo'lsa, m parametr qabul qilishi mumkin bo'lgan eng katta qiymatini toping.

A) 2 B) 4 C) -2 D) 6

11. Ifodani soddalashtiring.

$$\left(\left(\frac{m}{n}-\frac{n}{m}\right):(m+n)+m\left(\frac{1}{n}-\frac{1}{m}\right)\right):\left(1+\frac{m}{mn}\right)$$

A) 1 B) m − n C) m + n D) 2mn

12. Hisoblang.

$$\operatorname{ctg}\left(\frac{1}{2}\arcsin\frac{4}{5}\right)$$

A) $\frac{1}{2}$ B) $\frac{3}{2}$ C) 2 D) 3

13. Tengsizlikning $[-\pi;\pi]$ oraliqdagi eng katta va eng kichik yechimlari yig'indisini toping.

$$2\cos^2 x+5\cos x+2\geq 0$$

A) $-\frac{\pi}{3}$ B) $\frac{\pi}{3}$ C) 0 D) $\frac{2\pi}{3}$

14. Tenglamaning haqiqiy yechimlari yig'indisini toping.

$$3\cdot 9^{x+1}-6\cdot 3^x-1=0$$

A) -3 B) -1 C) 6 D) 0

15. Tenglamaning butun yechimlari ko'paytmasini toping.

$$\log^2_{0,5}(4x)+\log_2\frac{x^2}{8}=8$$

A) 1 B) 2 C) 16 D) 24

16. Tenglamaning haqiqiy yechimlari sonini toping.

$$|x^2+3|+|x^2-x+1|=5$$

A) 1 ta B) 2 ta C) 3 ta D) Yechimga ega emas

6. Hisoblang.

$$\frac{572}{(\sqrt{6}+\sqrt{7}+\sqrt{8})\cdot(\sqrt{6}+\sqrt{7}-\sqrt{8})\cdot(\sqrt{6}-\sqrt{7}+\sqrt{8})\cdot(-\sqrt{6}+\sqrt{7}+\sqrt{8})}$$

A) 1 B) 4 C) $2\sqrt{6}$ D) $2\sqrt{7}$

7. Soddalashtiring:

$$\frac{\sqrt[3]{343\cdot x^{\frac{9}{4}}\cdot\sqrt{250}}}{10}$$

A) $\frac{7\sqrt[3]{\sqrt{2x^3}}}{10}$ B) $\frac{7\sqrt[3]{\sqrt{4x^4}}}{10}$ C) $\frac{7\sqrt[3]{\sqrt{4x^9}}}{10}$ D) $\frac{7\sqrt[3]{\sqrt{4x^3}}}{5}$

8. Arifmetik progressiyaning 4-hadi 16 ga, 7-hadi va 10-hadi yig'indisi 5 ga teng bo'lsa, progressiyaning dastlabki 18 ta hadi yig'indisini toping.

A) -18 B) -9 C) -6 D) -12

9. Agar cheksiz kamayuvchi geometrik progressiyaning dastlabki ikkita hadi yig'indisi 24 va progressiyaning maxraji $\frac{1}{2}$ ga teng bo'lsa, progressiyaning barcha hadlari yig'indisini toping.

A) 48 B) 32 C) 50 D) 45

10. Agar a = $\sqrt[3]{129}$ va b = $\sqrt[12]{12}$ bo'lsa,

$$\left(\frac{a\sqrt{a}+b\sqrt{b}}{\sqrt{a}+\sqrt{b}}-\sqrt{ab}\right)\cdot\left(\frac{\sqrt{a}+\sqrt{b}}{a-b}\right)^2$$

ning qiymatini toping.

A) $2\sqrt[3]{129}$ B) $\sqrt{12}$ C) 1 D) 0